Boustead Cuping AI Surveillance Platform

Commercial Proposal Write-Up for 400-Acre Maize Farm Security

Prepared as a proposal basis from the working clickable prototype.

Scope note: satellite layer is real; boundary and zones are representative pending official parcel KML/GeoJSON and site confirmation.

1. Executive Summary

Boustead Cuping AI Surveillance is proposed as an integrated farm protection platform for a 400-acre maize operation. The platform is designed to detect, validate, and escalate the primary risks identified for the farm: wild boar incursion, human intrusion, arson intent, and dry-weather fire.

The solution is positioned as an operating system for security and farm operations, not only a camera-viewing dashboard. It combines perimeter map awareness, AI detections, camera evidence, incident workflow, drone response, device and firmware monitoring, automation rules, APIs, audit logs, and zero-hour support escalation.

AI is treated as a mandatory detection and prioritisation layer. Final enforcement or dispatch decisions should remain operator validated, especially for intrusion and arson-related events.

2. Site And Threat Context

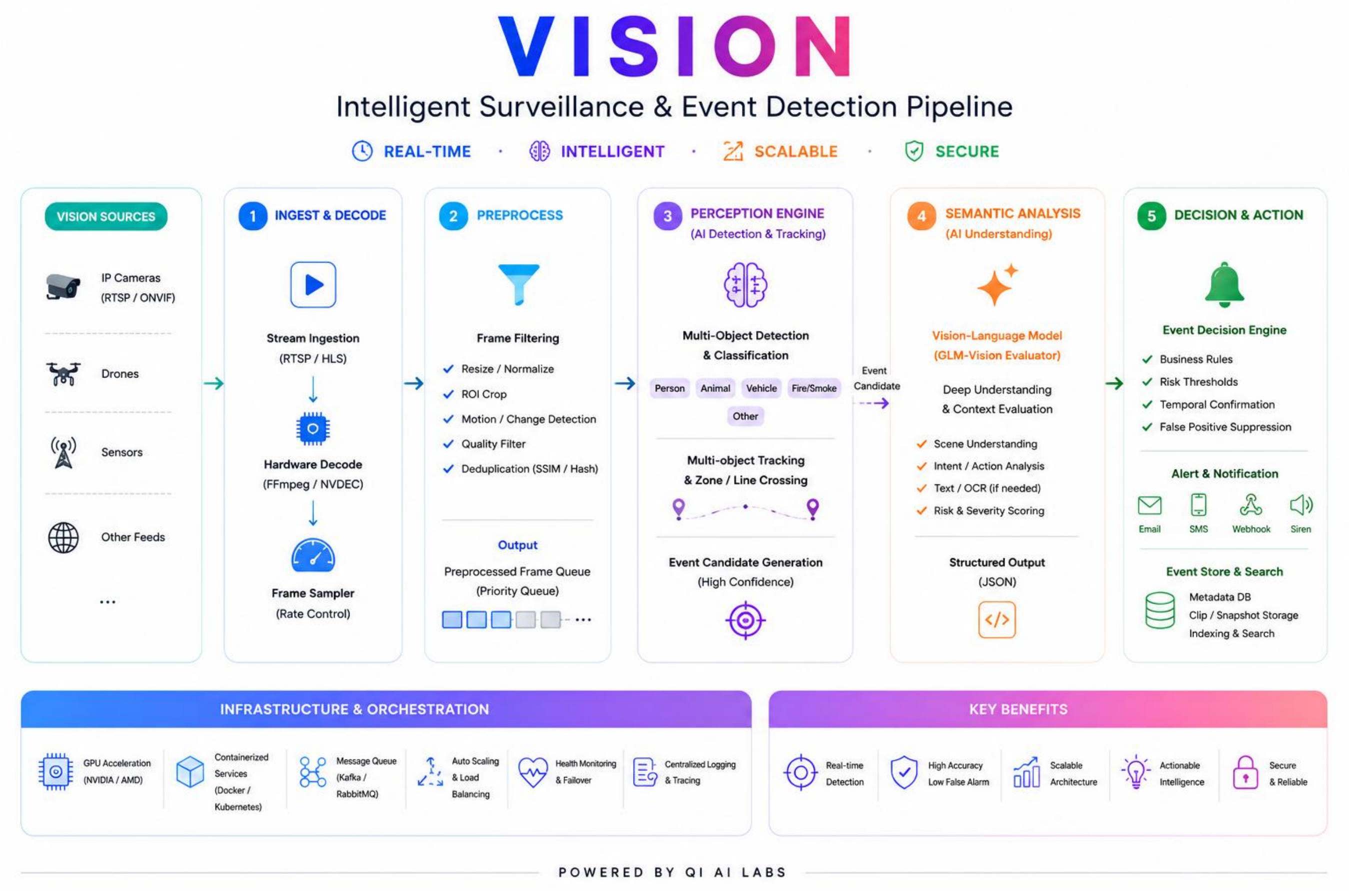
- **Planning baseline**: 400 acres of maize/corn farm land at Cuping farm land.
- **Primary threats**: wild boars, human intrusion, arson intent, and dry-weather fire.
- **Operational requirement**: surveillance, perimeter monitoring, camera feed integration, AI detection, incident escalation and evidence capture.
- **Data caveat:** the prototype uses representative zone boundaries. Final solution requires official parcel KML/GeoJSON, camera locations, tower locations, connectivity survey and response SOP confirmation.

3. Proposed Solution Overview

The proposed platform provides six operating capabilities:

- **Command Dashboard** - central KPIs, open incidents, priority queue, camera evidence wall, detection stream and operational status.
- **Zone Map** - real satellite map layer, representative 400-acre boundary, perimeter sectors, operating zones and animated zone focus.
- **Camera Evidence -** AI-labelled camera cards with threat type, confidence, model, latency, and evidence capture actions.
- **Incident Workflow** - lifecycle from new detection to acknowledgement, verification, dispatch and closure.
- **Drone Response** - drone launch, return-to-dock, route, battery, ETA and mission state for rapid verification.
- **Rules & API Configuration** - AI thresholds, schedules, response policies, module configuration, API endpoints, webhook retry and audit log.

4. Available Product Features



Feature Area	Available Functionality
AI Threat Detection	Wild boar, human intrusion, smoke/fire and arson-risk model families with confidence, severity and source device metadata.
Map Operations	Satellite map, 400-acre boundary overlay, perimeter sectors, risk zones, selected-zone zoom and local intelligence panel.
Focused Camera Feeds	Zone-filtered mock CCTV evidence cards with detection boxes, status, confidence, FPS, latency and model version.
Incident Workflow	Incident queue, evidence link, operator notes, state progression and feedback tagging.
Drone Operations	Drone launch, return, route chips, ETA, mission state, battery and AI tasking.
Device & Firmware	Gateway health, firmware, heartbeat, storage, network RSSI, tamper state, OTA, reboot and recovery action.
Rules & API	Confidence thresholds, detection duration, schedule, action policy, escalation, endpoint status, response matrix and audit trail.
Zero-Hour Support	High-severity support escalation flow for fire, arson, intrusion and dry-weather operating mode.

5. AI Rules And Module Configuration

The Rules & API module is intended to make AI behaviour commercially controllable. Instead of treating AI as a black box, each threat class can be configured by confidence threshold, detection duration, active zones, schedule, action policy and escalation level.

- Wild Boar: suitable for north ridge and harvest belt patrol escalation, especially during evening and early morning windows.
- Human Intrusion: suitable for gate, access road and central corridor escalation with security dispatch policy.
- Smoke / Fire: suitable for all zones, dry-weather preset and zero-hour callout policy.
- Arson Risk: suitable for night movement patterns, suspicious stopping behaviour and supervisor review before escalation.

Module configuration also covers evidence retention, pre-roll/post-roll clips, minimum FPS, inference mode, model update strategy, incident SLA, drone battery policy, wind limit, webhook retries, signed payloads and offline buffer.

6. Operator Workflow

- AI detects a potential threat from camera, thermal or sensor feed.
- The platform creates an event with threat type, zone, source device, confidence, model and recommendation.
- The event is mapped to an incident and appears in dashboard, zone map, incident queue and event stream.
- Operator reviews camera evidence, labels true/false positive, adds notes and verifies the event.
- If needed, operator dispatches ground team or launches drone verification.
- Incident is closed with evidence, feedback and audit record retained.

7. Technical Architecture Direction

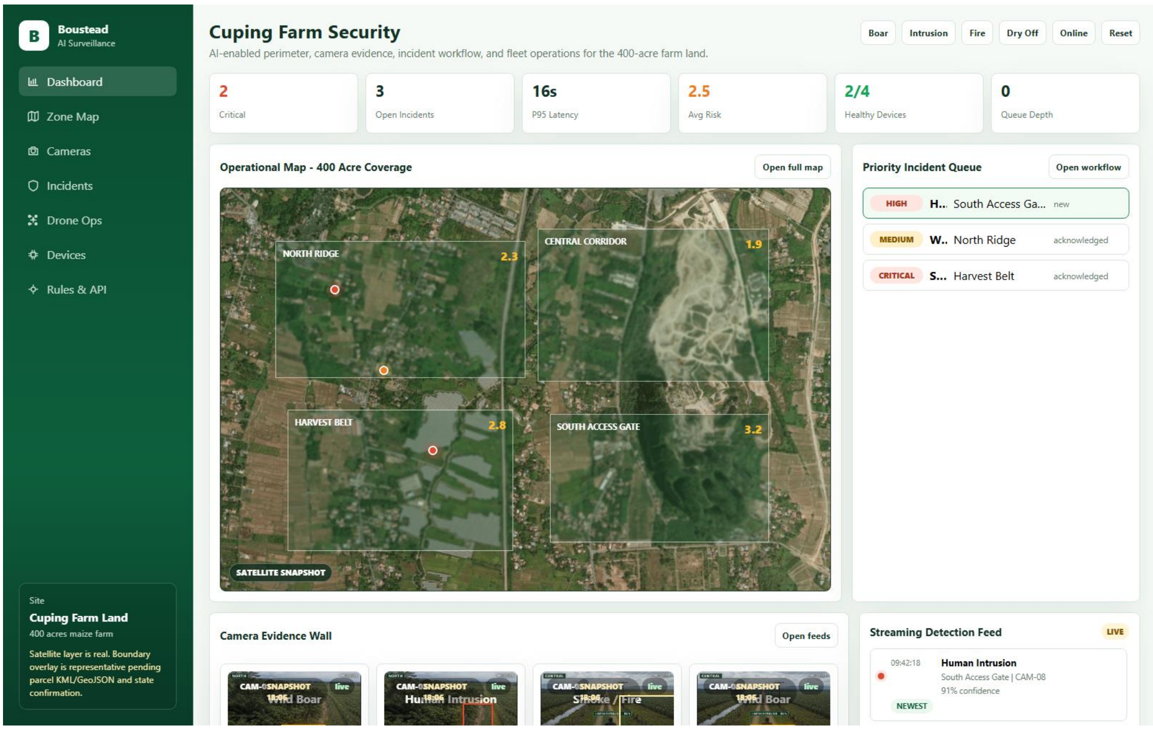
Recommended baseline architecture is hybrid edge-first: run real-time inference close to the camera gateway where possible, buffer events locally during connectivity loss, and sync event metadata, clips and audit records to a central backend. This reduces bandwidth dependency and improves response latency.

- Edge layer: camera ingestion, on-device inference, frame sampling, clip pre-roll/post-roll, health watchdog and local queue.
- Platform layer: event API, incident workflow, rules engine, audit log, webhook delivery, user roles and reporting.
- Operations layer: dashboard, zone map, camera evidence, drone response, firmware console and rules configuration.
- Integration layer: APIs for events, incidents, device health, sync status, drone launch and rule evaluation.
- Security layer: signed payloads, role-based access, audit logging, OTA controls and tamper/health monitoring.

8. Proposal Screenshots

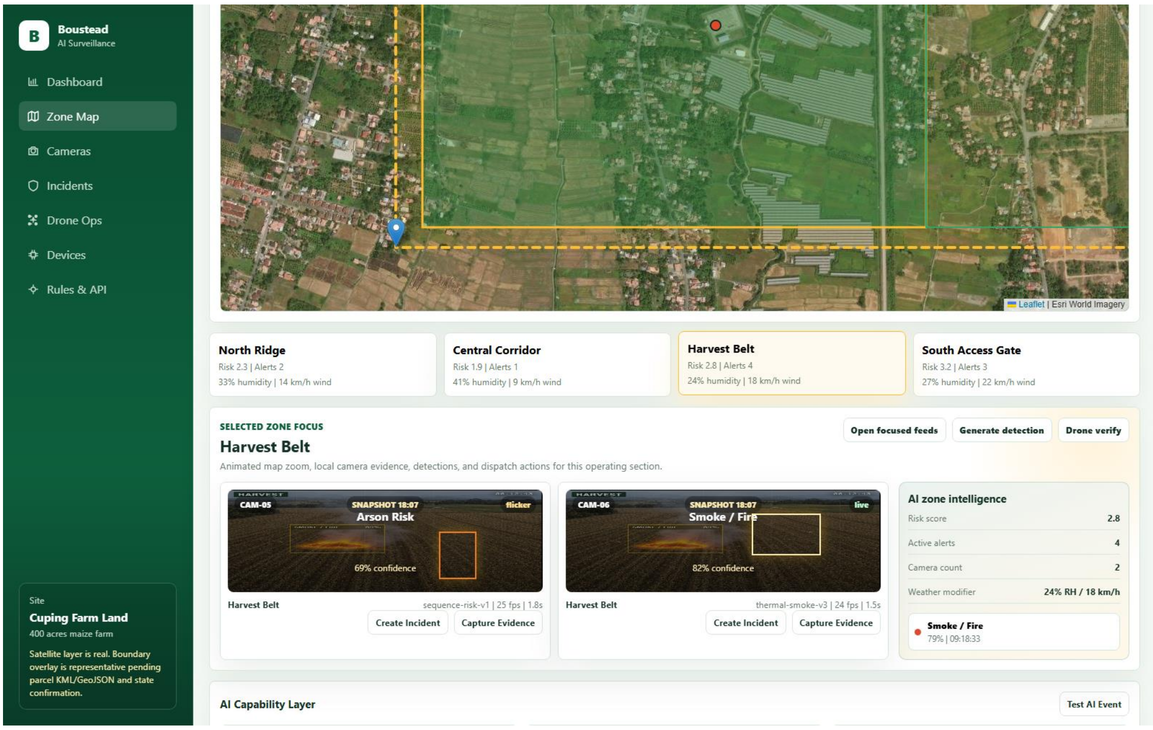
The following visuals are captured from the working prototype and can be reused in a proposal deck or RFI pack.

Figure 1. Command Dashboard



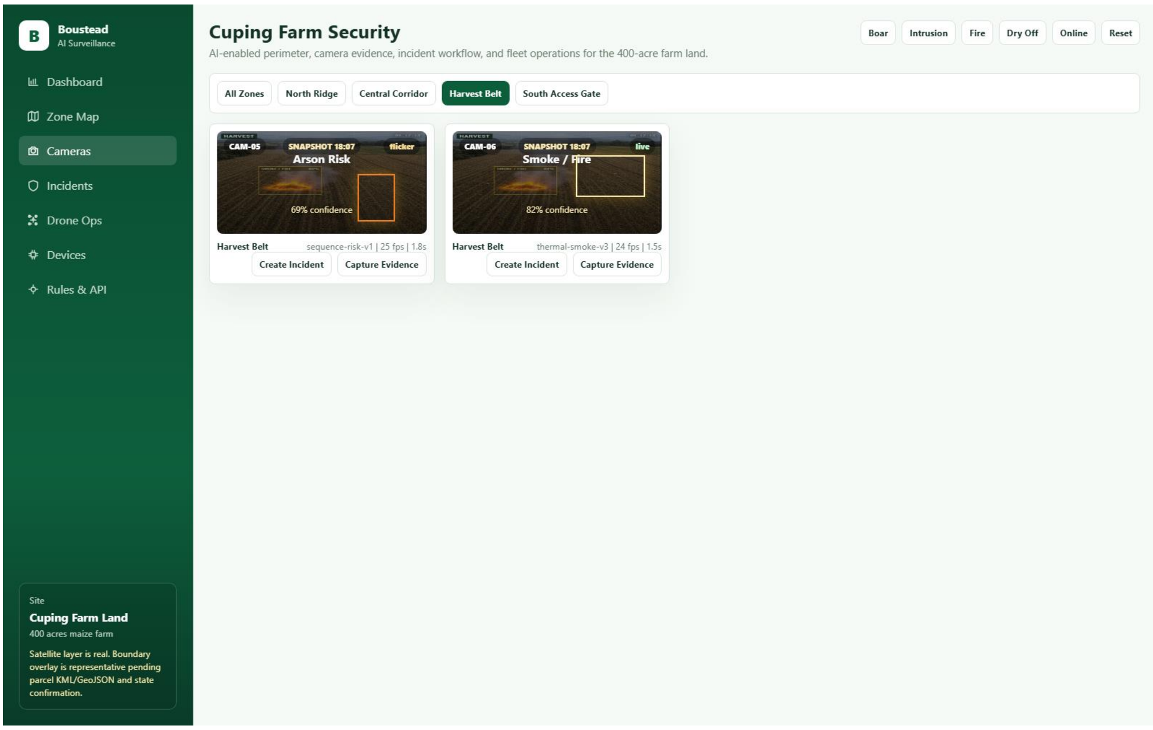
Command dashboard showing KPIs, priority incidents, camera evidence and detection stream.

Figure 2. Zone Focus Map



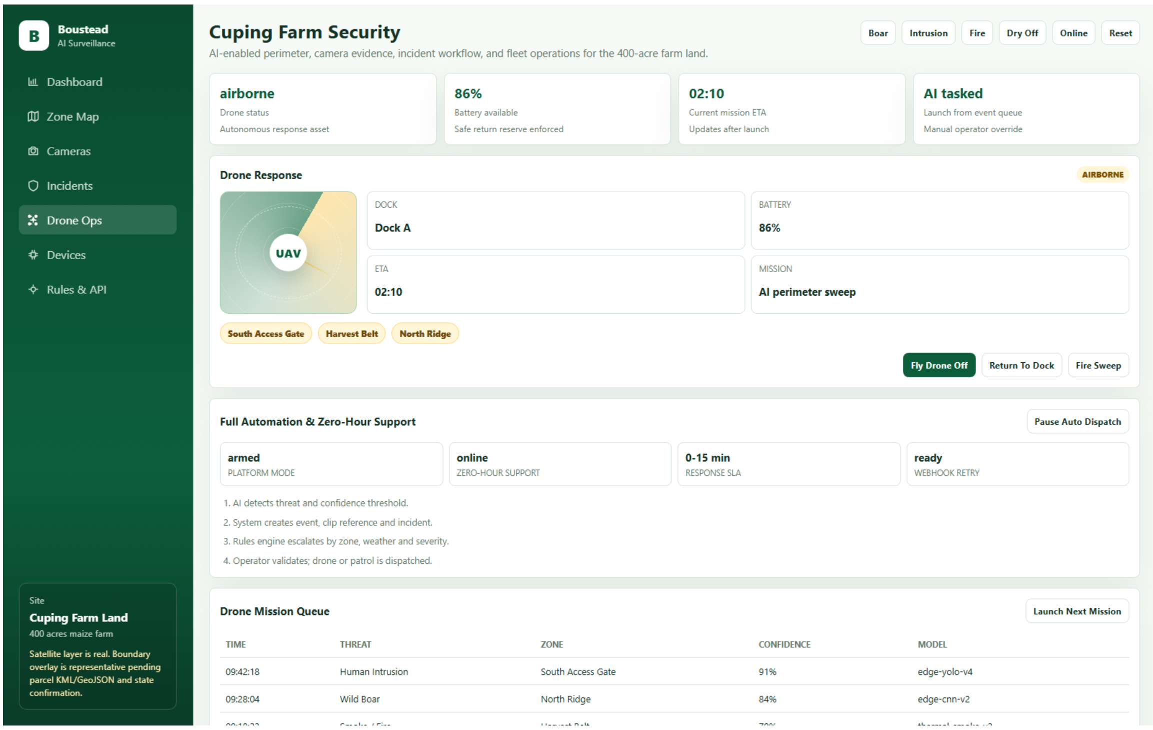
Zone map with representative farm boundary, selected Harvest Belt focus and local camera evidence.

Figure 3. Focused Camera Feeds



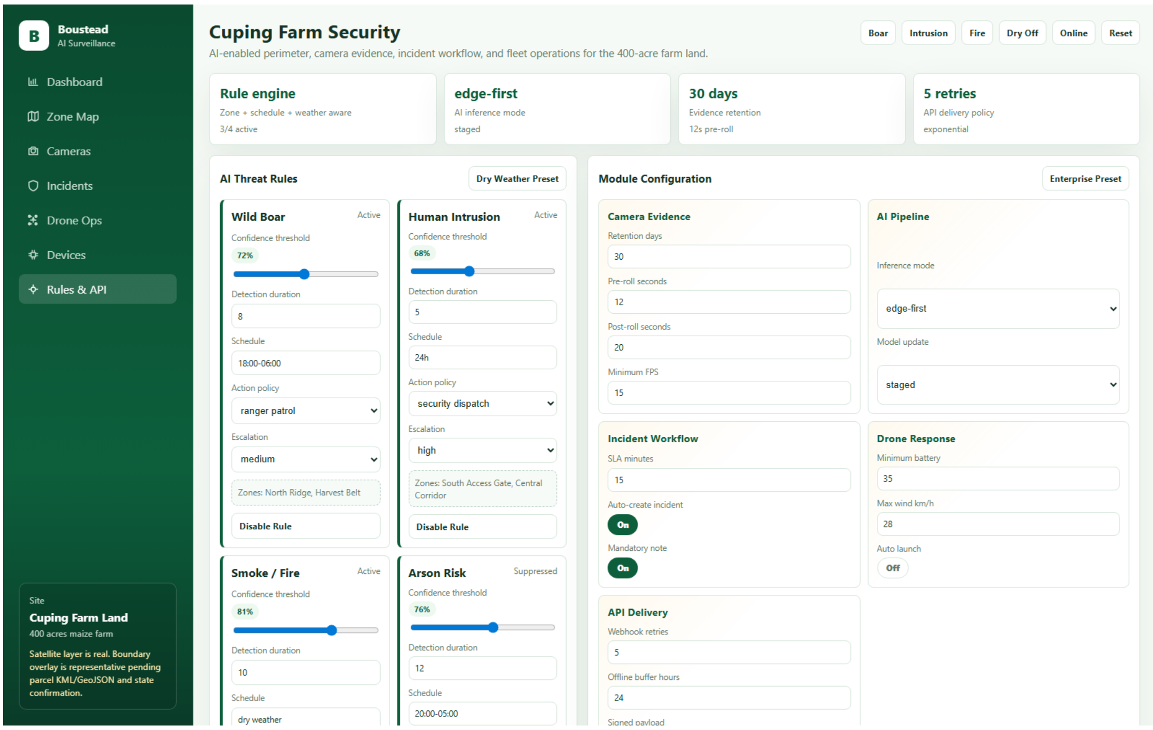
Focused camera feed screen filtered to the selected operating zone.

Figure 4. Drone Response



Drone response module showing mission state, battery, ETA, route and dispatch actions.

Figure 5. Rules and API Configuration



Rules and API module showing AI thresholds, module configuration, response matrix and audit trail.

9. Pilot Blueprint

Recommended pilot duration: 6 to 8 weeks across one or two high-risk zones, such as Harvest Belt and South Access Gate.

- Week 1-2: confirm parcel boundary, camera locations, power, connectivity and response SOP.
- Week 2-4: configure camera feeds, AI rules, event payload, incident workflow and device health monitoring.
- Week 4-6: run controlled event simulations for boar, intrusion, smoke/fire and arson-risk scenarios.
- Week 6-8: measure detection latency, false positive rate, missed detections, validation time, uptime and support response.

10. Commercial Deployment Tiers

Tier	Description	Commercial Use
Minimal Viable	Camera feed integration, dashboard, zone map, basic AI detection and incident workflow.	Pilot and proof of value.
Production Rollout	Expanded camera sectors, device health, OTA, rules engine, APIs, evidence retention and reports.	Full farm operations.
Resilient / Enterprise	Hybrid edge, thermal additions, drone response, redundant connectivity, zero-hour support and advanced audit.	Mission-critical security operation.

11. Key Risks And Validation Items

- Final parcel boundary and state/location confirmation are required before final coverage design.
- Camera placement must be validated against vegetation, crop growth, night visibility, rain, dust, fog and blind spots.
- AI model accuracy must be measured by local site data, not assumed from generic demonstrations.
- Drone operation requires legal, safety, weather and SOP validation before production use.
- Connectivity and power reliability must be surveyed for all towers, gateways and edge devices.
- Cybersecurity, access control and audit requirements must be confirmed before production deployment.

12. Conclusion

The proposed solution gives Boustead a commercially credible path from surveillance requirement to managed farm security operations. The strongest next step is to validate the exact land boundary, confirm the highest-risk zones, and run a 6-8 week pilot with measurable AI, incident and operational KPIs.